Q585 G1: Purbhoo's Proposition 5.1 reduced to an order-reversal statement

PROVE 2026-11-14 c1 — *not proved*; reduction proved, gap isolated

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Provenance (PROTOCOL §2.3)

- Code: `proofs/code-1114-Q585-G1/`, commit `625de42e5ab6` (already committed when this line was written). Engine: `code-1112c2-Q585/` and `code-1113-Q585-H1/readouts.py`, unchanged.
- Source: Purbhoo, *Puzzles, tableaux and mosaics*, arXiv `0705.1184`, local seed PDF. Statement read first-hand (pdftotext, printed p. 19, §5.1). Figure 9 rendered at 110 dpi (printed p. 20). §4 (proofs of Theorems 1, 2, Wake Crossing Lemma) read first-hand (printed pp. 17–18).
- Status: G1 **not proved**. Theorem 6 (a reduction) **is** proved, modulo Purbhoo's published Theorem 1, Lemma 2.6, and the mosaic–puzzle bijection.

1 The statement

Verbatim (printed p. 19): *"Proposition 5.1. The following bijections between mosaics and Littlewood-Richardson tableaux coincide: (i) Tao's bijection as formulated in Figure 9; (ii) the bijection obtained by α orientation (E_A, N_A) and migrating from A to B (second bijection in Table 1); (iii) the bijection obtained by giving α orientation $(-E_A, -N_A)$ and migrating from A to C."* The preamble to §5.1 says: *"We will omit proofs here, since they are long but relatively straightforward."*

Write $R_{\mathrm{ii}}(M) = \mathcal{R}_A(+, N)(M)$ for (ii) and $R_{\mathrm{iii}}(M) = \mathcal{R}_A(-, W)(M)$ for (iii). In the frame $(-E_A, -N_A)$, the direction W is $+E_A$, which points towards C. Both readouts produce an LR tableau in the destination nest, recorded in the orientation that Theorem 1 induces there.

Remark 1 (Leg (i) is untested)**.** In every earlier session, the engine's "Tao" is literally `readout(b,0,1,'N')`, which is R_{ii}. Figure 9 is a picture with no algorithm attached. So leg (i) has never been implemented independently, and nothing below concerns it. G1 as used downstream (G2, G3, H1) needs only $R_{\mathrm{ii}} = R_{\mathrm{iii}}$.

2 Computation, with controls (Step 0)

- **The small census is zero bits.** If the fibre (the mosaics with a fixed boundary) has size 1, then any two bijections onto the LR fibre agree. Over *all* KT puzzles with $n \le 5$ there is no fibre of size ≥ 2: the 146/146 agreements carry nothing. For $n = 6$ there is exactly one

such fibre (2 mosaics). For $n = 7$ (all puzzles, mosaics built directly from puzzles) there are 21 fibres, and $R_{\mathrm{ii}} = R_{\mathrm{iii}}$ holds on 42/42.

- $c \geq 2$ **census** (mosaics built by inverting Cor. 3.2; $|\lambda|, |\mu| \leq 5$; 67 cells): $R_{\mathrm{ii}} = R_{\mathrm{iii}}$ holds on 134/134.
- **Controls on the same fibre.** C3 (the R_{ii} value of the *next* mosaic in the fibre) differs from R_{iii} on 134/134 and 42/42. C1 $= \mathcal{R}_A(+, E)$ and C2 $= \kappa = \mathcal{R}_A(-, S)$ occupy the same cells as R_{ii} on only 12/134 (18/42) elements; on those they differ every time. So the brief's κ-lesion fires mostly by *shape* mismatch, which is cheap. The fibre-shift C3 is the informative control.
- **Tie-break.** Every $c \geq 2$ element meets the flagged minimal-hexagon tie-break (2212 ties over 134 elements). With the tie filter *inverted*, migration fails to terminate on 134/134. The flagged reading is therefore the only one of the two that defines a map on this census.

3 The reduction

Let M have boundary (α, β, γ) with $m = |\alpha|$. In what follows, the *standard order* on a flock means: $\diamond < \diamond'$ if $f(\diamond) < f(\diamond')$, or if the values are equal and $\diamond$ is west of $\diamond'$ (Purbhoo §2.3).

Lemma 2 (the two unique A-flocks)**.** *Index the cells of α by (c, r) in the frame (E_A, N_A), with row r of length α_{r+1}, and let α'_c be the height of column c.*

1. *In the orientation (E_A, N_A), the unique flock (Lemma 2.5) is $f_+(c, r) = r + 1$.*
2. *In the orientation $(-E_A, -N_A)$, where α appears rotated by $180°$, the unique flock is $f_-(c, r) = \alpha'_c - r$ (the depth of the cell, counted from the top of its column).*

Proof. By Lemma 2.5 each flock is unique, so it suffices to check that each filling is an LR tableau with content α. (1) This is the standard Yamanouchi filling. (2) Content: the number of k's is $\#\{c : \alpha'_c \geq k\} = \alpha_k$. Columns: in the rotated frame, north is $-N_A$, i.e. decreasing r, and f_- strictly increases as r decreases. Rows: east in the rotated frame is west in the original, and α'_c is weakly decreasing in c, so f_- is weakly increasing eastward. Lattice condition: the reversed reading word in the rotated frame reads the original rows from $r = \ell(\alpha) - 1$ down to 0, each row west to east. After the rows $\geq r$ have been read completely, the number of k's read is $\#\{c : \alpha'_c \geq k + r\} = \alpha_{k+r}$, and this is weakly decreasing in k. Now suppose row r has been read only through column j. Then

$$\#k = \alpha_{k+r+1} + \#\{c \leq j : \alpha'_c = k + r\} \;\geq\; \alpha_{k+r+1} \;\geq\; \alpha_{k+r+2} + \#\{c : \alpha'_c = k + r + 1\} \;\geq\; \#(k+1),$$

using $\#\{c : \alpha'_c = i\} = \alpha_i - \alpha_{i+1}$. □

Computer check (pure, independent of the engine): 96/96 shapes α with $|\alpha| \leq 9$, both parts.

Corollary 3 (processing orders)**.** *(ii) migrates in direction N, so it processes from the maximum downwards. It therefore takes the top row of α first, east to west within a row, and then the next row down. (iii) migrates in direction W, so it processes from the minimum upwards in the rotated standard order. It therefore takes all the depth-1 cells (the column tops) first, east to west in the original frame, then the depth-2 cells, and so on. Every removal in either order is an outer corner of what remains.*

Lemma 4 (forcing)**.** *A mosaic whose A-nest is empty has boundary $(\emptyset, \gamma^\vee, \gamma)$ for some γ, and it is the unique mosaic with that boundary.*

Proof. Mosaics with boundary (α, β, γ) are in bijection with KT puzzles with the same boundary (Purbhoo §2.4), and these are counted by the symmetric structure constant $c_{\alpha\beta\gamma}$ [KT]. For $\alpha = \emptyset$ this constant is $\delta_{\beta,\gamma^\vee}$. Existence and uniqueness for $(\emptyset, \gamma^\vee, \gamma)$ is used by Purbhoo in the proof of Cor. 3.2 (p. 15) and is his Lemma 2.6. □

Lemma 5 (rigid transport; T2)**.** *Let N be a mosaic with empty A-nest, and let a rhombus at an outer corner b of the B-nest migrate and land in C. Then the landing box is $\iota(b)$, the box that is dual to b under complementation. The resulting mosaic is again the unique one from Lemma 4.*

Proof. After landing, the A-nest is still empty and the B-nest is $\gamma^\vee \setminus b$. By Lemma 4 the C-nest must be $(\gamma^\vee \setminus b)^\vee = \gamma \cup \iota(b)$. □

In the induced frames, ι is a translation: it is the same constant translation for every rhombus, on 134/134 elements. This frame bookkeeping is checked, not proved. We use ι to identify a tableau on β'/β in B with a tableau on γ'/γ in C. Because ι is a translation, it preserves the standard order.

Theorem 6 (reduction)**.** *Let $b_1, \ldots, b_m$ be the boxes of B in the order in which (ii) fills them, and $c_1, \ldots, c_m$ the boxes of C in the order in which (iii) fills them. Then*

$$R_{\mathrm{iii}}(M) = \iota_* R_{\mathrm{ii}}(M) \iff c_k = \iota(b_{m+1-k}) \ \text{ for all } k \qquad \textit{(Order Reversal).}$$

Proof. By Purbhoo's Theorem 1, migration preserves the standard order, and migration carries the values. In (ii) the rhombi are processed in decreasing standard order (Cor. 3). Hence $b_1, \ldots, b_m$ is the decreasing standard order of $R_{\mathrm{ii}}(M)$, and $R_{\mathrm{ii}}(M)(b_k) = v_k$, where $v = (\ell^{\alpha_\ell}, \ldots, 2^{\alpha_2}, 1^{\alpha_1})$ by Lemma 2(1). In (iii) the rhombi are processed in increasing standard order. Hence $c_1, \ldots, c_m$ is the increasing standard order of $R_{\mathrm{iii}}(M)$, and $R_{\mathrm{iii}}(M)(c_k) = w_k$, where $w = (1^{\alpha_1}, 2^{\alpha_2}, \ldots)$ by Lemma 2(2). Note that $w_k = v_{m+1-k}$.

($\Leftarrow$) We have $R_{\mathrm{iii}}(c_k) = w_k = v_{m+1-k} = R_{\mathrm{ii}}(b_{m+1-k}) = (\iota_* R_{\mathrm{ii}})(c_k)$.

($\Rightarrow$) If $R_{\mathrm{iii}} = \iota_* R_{\mathrm{ii}}$, the two tableaux have the same increasing standard order, because ι preserves the standard order. So c_k is the k-th smallest box of $\iota_* R_{\mathrm{ii}}$, which is $\iota(b_{m+1-k})$. □

Remark 7 (what carries the bits)**.** By Lemma 2.6, the *final mosaic* of either readout is forced by the boundary; it carries no information. The *values* are forced by the order (Theorem 6). So the whole content of Prop. 5.1 (ii)=(iii) is a statement about *order across two different migrations*. Purbhoo's Theorem 1 (the Wake Crossing Lemma) controls order only *within* one migration.

Computer check of the reduction. On the $c \geq 2$ census: the arrivals of (ii) are the decreasing standard order of R_{ii} (134/134); the arrivals of (iii) are the increasing standard order of R_{iii} (134/134); and Order Reversal holds (134/134). The unreversed control $c_k = b_k$ holds 0/134.

4 Attempts on Order Reversal

1. **Factorisation (T1), computed.** Let (ii) land its flock F' in B. Migrating F' onwards from B in direction S of its induced frame lands in C, and reproduces R_{iii}: the tableau and the frame agree on 134/134. The other three directions get stuck, loop, or return to A. With Lemma 5, T1 is *equivalent* to G1, so it does not reduce the gap.

2. **Single-rhombus coherence (the induction route), refuted.** For each outer corner ρ of α, compare sending ρ A→B and then B→C with sending it A→C directly. They agree on 0/268 pairs (all 134 mosaics, all corners). So T1 holds only for whole flocks with A emptied, and induction on $|\alpha|$ one rhombus at a time is dead.

3. **Inverse lockstep (Theorem 2).** Order Reversal says: R_{iii}^{-1} (C→A, maximum first) empties C in exactly the order in which (ii) fills B, transported by ι. Equivalently, for every k, the C-nest of the k-th intermediate mosaic of R_{iii}^{-1} is the dual of the B-nest of the k-th intermediate mosaic of (ii). The two intermediate mosaics have A-nests of sizes $m-k$ and k, and their removal orders are different (rows versus column tops). So no mosaic-level identity pairs them, and I did not find an invariant that does.

Remark 8 (two equivalent forms of the gap)**.** (a) *FIFO form.* By Theorem 2, the inverse of (ii) sends rhombi back in reverse order. So Order Reversal says: under ι, the order in which R_{ii}^{-1} (B→A) *empties* B equals the order in which (iii) *fills* C. In other words, the composite B→A→C is first-in-first-out on *boxes*. It is not FIFO on *physical rhombi*: the cell $(0,0)$ of α enters A first and leaves it last (Cor. 3). So the order information travels through the deformation of the tiling, not with the tiles. (b) *Role in the H1 chain.* Yesterday's Lemma `lem:rot` (`2026-11-13-Q585-H1-migration-is-switching.tex`) gives $\Phi_G(P) = R_{\mathrm{iii}}(\rho^{-1}P)$. In the H1 chain, G1 is used only to turn this R_{iii} into R_{ii}, so that G3 (Cor. 5.3, which is stated for R_{ii}) applies. The composite *mixed* identity $R_{\mathrm{iii}}(\rho^{-1}P) = \Phi(R_{\mathrm{ii}}(P))$ is what H1 actually consumes. It was computed 134/134 on 11-13. A proof of that mixed identity would *bypass* G1.

5 Gap (precise)

Order Reversal. For every mosaic M, the box sequence in which $\mathcal{R}_A(+,N)$ fills B is, under ι, the reverse of the sequence in which $\mathcal{R}_A(-,W)$ fills C. It is computed (134/134 and 42/42) and not proved. Two candidate routes remain: (a) a two-migration Wake Crossing argument, comparing the midlines of the (ii) and (iii) journeys; (b) the jdt route Purbhoo indicates (Prop. 5.2: a single B→C rhombus migration is a Schützenberger slide, proof omitted), which would turn Order Reversal into a statement about the reversal of standard order under jdt.

References

[P] K. Purbhoo, *Puzzles, tableaux and mosaics*, arXiv:0705.1184 (local seed PDF). Prop. 5.1 (p. 19), Fig. 9 (p. 20), Thm. 1/2 and the Wake Crossing Lemma (pp. 14, 17–18), Lemma 2.5 and 2.6, Table 1 (p. 15).
[KT] A. Knutson, T. Tao, *Puzzles and (equivariant) cohomology of Grassmannians* (recalled; puzzle count = $c_{\alpha\beta\gamma}$, as calibrated by the engine on 340/340 cases, registry node `q585-puzzle-calibration`).